

JIAYU ZHANG

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Education

Columbia University, Ph.D. in Operations Research, Aug. 2024 - **Present**

- Advisor: Tianyi Lin (<https://ieor.columbia.edu/content/tianyi-lin>)

Columbia University, M.S. in Operations Research, Aug. 2024 - Oct. 2025

Shanghai Jiao Tong University, B.S. in Mathematics, Aug. 2020 - Jun. 2024

Publications

Preprints

- Goldfarb, Donald, Lexiao Lai, Tianyi Lin, and Jiayu Zhang. "Non-Convex Self-Concordant Functions: Practical Algorithms and Complexity Analysis." arXiv preprint arXiv:2511.15019 (2025).
- Lei, Mingyu, Jiayu Zhang, and Yinyu Ye. "Blessing of high-order dimensionality: from non-convex to convex optimization for sensor network localization." arXiv preprint arXiv:2308.02278 (2023).

Undergraduate Thesis

- Zhang, Jiayu. "Accelerated Gradient Descent with Long Step-Size." Shanghai Jiao Tong University (2024). Supervisor: Jingwei Liang.
 - Automated the formulation of Performance Estimation Problems for Nesterov's Accelerated Gradient and Heavy-Ball Methods in Julia (JuMP with Knitro/Ipopt/Gurobi) and achieved $\sim 0.7x$ tighter computer-verified convergence bounds by computing and using optimal step-sizes and momentum factors.

Experiences

Cardinal Operations, Student Researcher, Feb. 2023 - Sep. 2023, Shanghai, China

- Achieved a $\sim 10x$ speedup in large-scale sensor network localization by implementing a stochastic gradient-based solver, outperforming traditional semidefinite programming baselines.

- Developed Python/MATLAB benchmarking pipelines for up to 10,000 nodes, evaluating commercial and custom solvers across varying network sizes, sensor densities, and noise levels.
- Co-authored an arXiv paper on algorithmic performance; developed theoretical proofs showing how high-dimensional warm starts improve optimization geometry in nonconvex landscapes.

Teaching

Columbia University, Teaching Assistant

- Fall 2025: IEOR 6613 Optimization I (Ph.D. level), instructor: Vineet Goyal.
- Spring 2025: IEOR 4004 Optimization Models and Methods I (graduate level), instructor: Vineet Goyal.

Fellowships and Awards

- Fall 2024: Boyle Fellowship, Columbia University.
- Fall 2022: Excellent Student Scholarship, Shanghai Jiao Tong University.

Skills

- Coding: Python, Julia, MATLAB, R; PyTorch, JuMP; Linux, Git, Bash, Slurm, LaTeX.
- Languages: English (proficient), Mandarin (native).